```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```

[illegible]

// 1NF → 2NF → 3NF [illegible]

1NF [illegible]	**2NF** [illegible]	**3NF** [illegible]	[illegible] OLAP [illegible]

$ cat schema-plan.md

□□□□

01 / □□

□□□□□□□□□□

02 / □□□□

1NF □□□ → 2NF → 3NF

03 / □□

JOIN □□□

04 / □□□□

OLTP vs OLAP □□□

□□□□□□□□□□□ —— □□□□□□□□□

PART 01

[illegible]

[illegible] N [illegible]

□□□□□□□□□□□□□

order_id	customer_name	customer_email	product_name	price
1	□□	zhang@example.com	□□□□□	30000
2	□□	zhang@example.com	□□	500
3	□□	li@example.com	□□	1200

□□□□
□□□□□□ email □□□□

□□□□
□□□ email□□ UPDATE □□□□

→

ROOT CAUSE

CHECKPOINT_01

[illegible] UPDATE 37 [illegible] 3 [illegible]

- A [illegible]
- B [illegible]
- C [illegible]

PART 02

□□□□

1NF → 2NF → 3NF□□□□□□□

1NF [illegible]

BEFORE — [illegible]

student	courses
[illegible]	[illegible], [illegible], [illegible]

AFTER — 1NF

student	course
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

[illegible]

[illegible]

[illegible]

2NF [illegible]

[illegible] (order_id, product_id) [illegible] product_name [illegible] product_id —— [illegible]

BEFORE — [illegible]

order_id	product_id	product_name	qty
1	P1	[illegible]	1
2	P1	[illegible]	3

AFTER — [illegible]

order_id	product_id	qty
1	P1	1

product_id	product_name
P1	[illegible]

[illegible]

3NF [illegible]

student_id → dept_id → dept_name [illegible] dept_name [illegible] dept_id [illegible] —— [illegible]

BEFORE — [illegible]

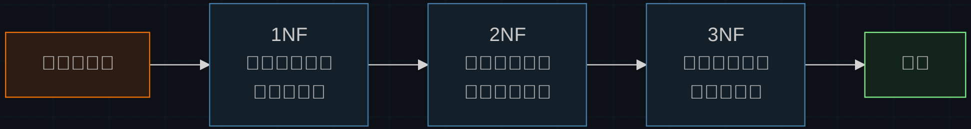
student_id	name	dept_id	dept_name
S1	[illegible]	D1	[illegible]
S2	[illegible]	D1	[illegible]

AFTER — [illegible]

student_id	name	dept_id
S1	[illegible]	D1

dept_id	dept_name
D1	[illegible]

[illegible] — [illegible]



3NF

PART 03

JOIN

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

JOIN [illegible]

[illegible]

[illegible]

[illegible] JOIN [illegible]

[illegible]

[illegible]

OLTP vs OLAP [illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]JOIN[illegible]	[illegible] JOIN
[illegible]	[illegible]	[illegible]
[illegible]	**OLTP** [illegible]	**OLAP** [illegible]

STRATEGY

[illegible] — [illegible]

[illegible]

[illegible]

[illegible] → [illegible]

[illegible]

1NF [illegible] → 2NF [illegible] → 3NF [illegible]

3NF [illegible]

[illegible]

[illegible]

OLTP [illegible]OLAP [illegible]

[illegible]

Schema normalized.

[illegible] JOIN [illegible]

```
$ mysql -e "EXPLAIN SELECT ..."
1NF → 2NF → 3NF → [illegible]
```